Machine Learning

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Lecture 10: Multiple Linear Regression

Solving System of Linear Equations using Matrix Algebra

- **Overdetermined System:** n > m
- **Underdetermined System:** n < m
- Possible Solutions for Square Matrix X:
  - Gaussian Elimination
  - Cramer's Rule Simple
  - Matrix Inversion Possible
- Solutions for Non-Square Matrix X:
  - Gauss Jordan Elimination
  - Moore Penrose Pseudo-inverse
  - SVD, QR Decomposition, and Cholesky Decomposition

$$A_{n\times m} \quad x_{m\times 1} = b_{n\times 1}$$

$$x = A^{-1} \quad b$$

$$X_{n\times m} \quad \beta_{m\times 1} = Y_{n\times 1}$$

$$\beta = X^{-1} \quad Y$$

Multiple Linear Regression

- An extension of simple linear regression that models the relationship between one dependent variable and multiple independent variables.
- To predict outcomes and understand the relationships between variables.

$$\boldsymbol{y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots\cdots + \beta_n x_n} + \in$$

- Explanation:
  - y: Dependent variable
  - x_1, x_2,…, x_n: Independent variables
  - β0: Intercept
  - β_1,β_2,…,β_n: Coefficients
  - ϵ\epsilon: Error term

$$\hat{y} = \beta_0 + \sum_{i=1}^{m} \beta_i x_i$$

Multiple Linear Regression (Example)

Dependent Variable (y)	Independent Variable (x_1)	Independent Variable (x_2)	Independent Variable (x_3)
GPA of a student	Number of hours studied daily	Teaching Aids	Instructor Qualification
Forgetness level	Drug dosage in ml	Social bindings	Family Environment
Salary of a person	Number of Education years	Managerial skills	Communication skills
House price	Covered area of the house	Number of bed rooms	distance from office
Electricity Bill	Amount of electricity consumed	Loadshedding	Billing slabs
Distance travelled	Time	type of car	traffic conditions
Sales of AC	Advertising Expenditures	Marketing	Season

Multiple Regression (Dataset)

	x1	x2	x3	y
0	89	4	3.84	7.0
1	66	1	3.19	5.4
2	78	3	3.78	6.6
3	111	6	3.89	7.4
4	44	1	3.57	4.8
5	77	3	3.57	6.4
6	80	3	3.03	7.0
7	66	2	3.51	5.6
8	109	5	3.54	7.3
9	76	3	3.25	6.4

Points to Ponder for Multiple Regression

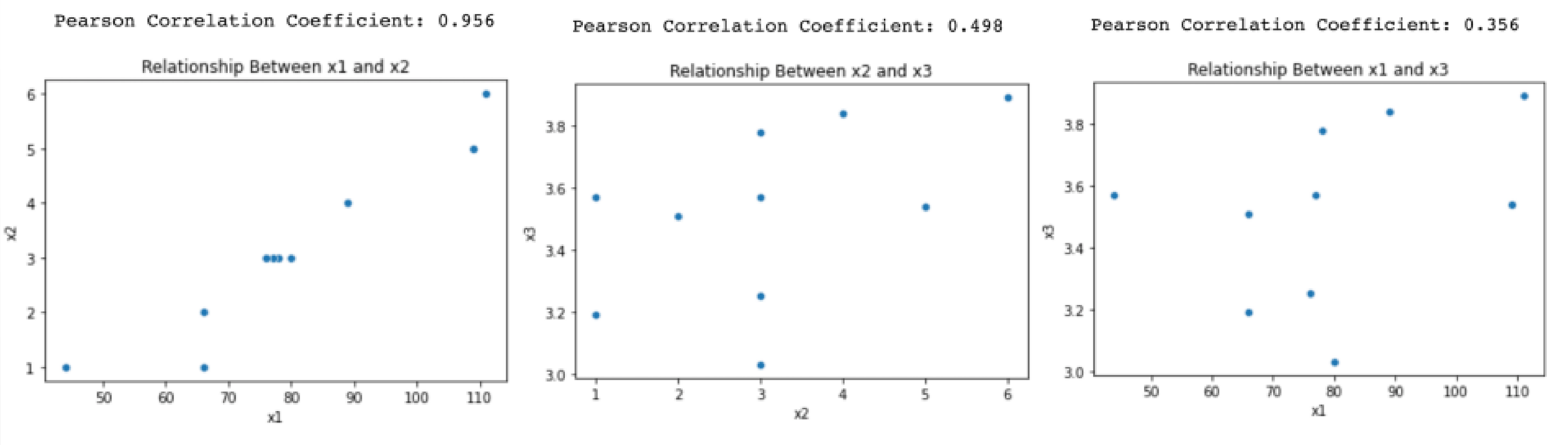
- Overfitting
- Multicollinearity

Overfitting

- Overfitting is caused by adding too many feature variables, which account for more variance but add nothing to the model.
- Multicollinearity occurs when some or all of the independent variables are correlated with each other.
- We can check the relationships between each independent variable and the dependent variable using scatter plot and correlations.
- Since there are three input variables, so we have three relationships to analyze (between three feature variables and the only output variable).
- Before using multiple linear regression, one must ensure that there should exist a linear relationship between the dependent variable (y), and all the independent variables (x , ,)

Multicollinearity

- Multicollinearity occurs when some or all of the independent variables are correlated with each other.
- Since there are three input or feature variables, so we have three relationships among them to analyze.
- All the feature variables should be somehow correlated with the output variable, but not with each other.



Formulation of Multiple Linear Regression using Matrix Representation

$$\hat{y} = \beta_0 + \sum_{i=1}^{m} \beta_i x_i$$

$$\begin{aligned}
\hat{y}_1 &= \beta_0 + \beta_1 X_{1,1} + \beta_2 X_{1,2} + \beta_3 X_{1,3} + \cdots + \beta_m X_{1,m} \\
\hat{y}_2 &= \beta_0 + \beta_1 X_{2,1} + \beta_2 X_{2,2} + \beta_3 X_{2,3} + \cdots + \beta_m X_{2,m} \\
\hat{y}_3 &= \beta_0 + \beta_1 X_{3,1} + \beta_2 X_{3,2} + \beta_3 X_{3,3} + \cdots + \beta_m X_{3,m}
\end{aligned}$$

Formulation of Multiple Linear Regression using Matrix Representation

- The above overdetermined system of linear equations with n equations and m+1 unknowns can be written in matrix form as:

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 X_{1,1} + \beta_2 X_{1,2} + \beta_3 X_{1,3} + \cdots + \beta_m X_{1,m} \\ \beta_0 + \beta_1 X_{2,1} + \beta_2 X_{2,2} + \beta_3 X_{2,3} + \cdots + \beta_m X_{2,m} \\ \beta_0 + \beta_1 X_{3,1} + \beta_2 X_{3,2} + \beta_3 X_{3,3} + \cdots + \beta_m X_{3,m} \\ \vdots \\ \beta_0 + \beta_1 X_{n,1} + \beta_2 X_{n,2} + \beta_3 X_{n,3} + \cdots + \beta_m X_{n,m} \end{bmatrix}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} 1 & X_{1,1} & X_{1,2} & X_{1,3} & \cdots & X_{1,m} \\ 1 & X_{2,1} & X_{2,2} & X_{2,3} & \cdots & X_{2,m} \\ 1 & X_{3,1} & X_{3,2} & X_{3,3} & \cdots & X_{3,m} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,1} & X_{n,2} & X_{n,3} & \cdots & X_{n,m} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}$$

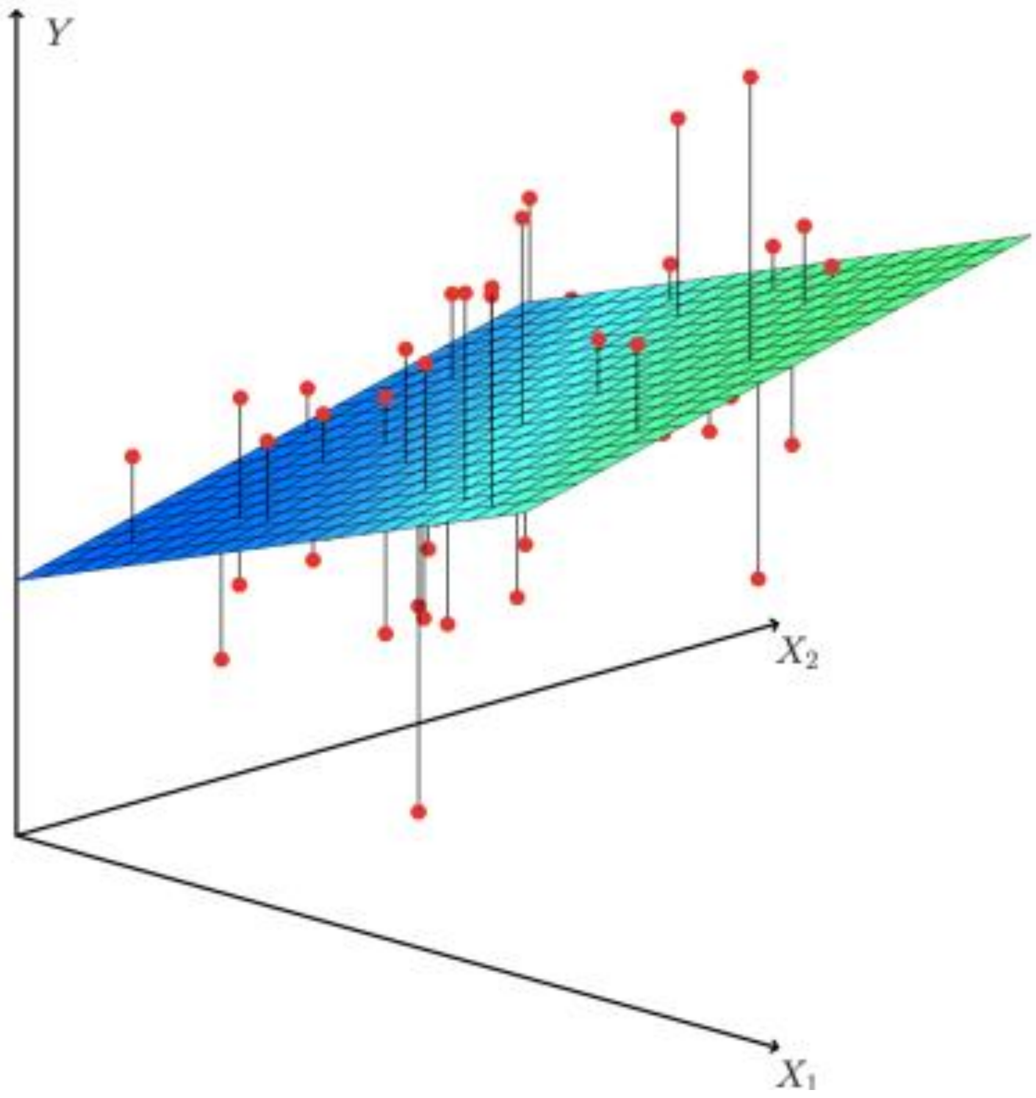
Formulation of Multiple Linear Regression using Matrix Representation

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 X_{1,1} + \beta_2 X_{1,2} + \beta_3 X_{1,3} + \cdots + \beta_m X_{1,m} \\ \beta_0 + \beta_1 X_{2,1} + \beta_2 X_{2,2} + \beta_3 X_{2,3} + \cdots + \beta_m X_{2,m} \\ \beta_0 + \beta_1 X_{3,1} + \beta_2 X_{3,2} + \beta_3 X_{3,3} + \cdots + \beta_m X_{3,m} \\ \vdots \\ \beta_0 + \beta_1 X_{n,1} + \beta_2 X_{n,2} + \beta_3 X_{n,3} + \cdots + \beta_m X_{n,m} \end{bmatrix}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} 1 & X_{1,1} & X_{1,2} & X_{1,3} & \cdots & X_{1,m} \\ 1 & X_{2,1} & X_{2,2} & X_{2,3} & \cdots & X_{2,m} \\ 1 & X_{3,1} & X_{3,2} & X_{3,3} & \cdots & X_{3,m} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,1} & X_{n,2} & X_{n,3} & \cdots & X_{n,m} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}$$

$$\hat{Y}_{n\times 1} = X_{n\times(m+1)} \quad \beta_{(m+1)\times 1}$$

Intuitive Understanding of the Cost Function for Multiple Linear Regression



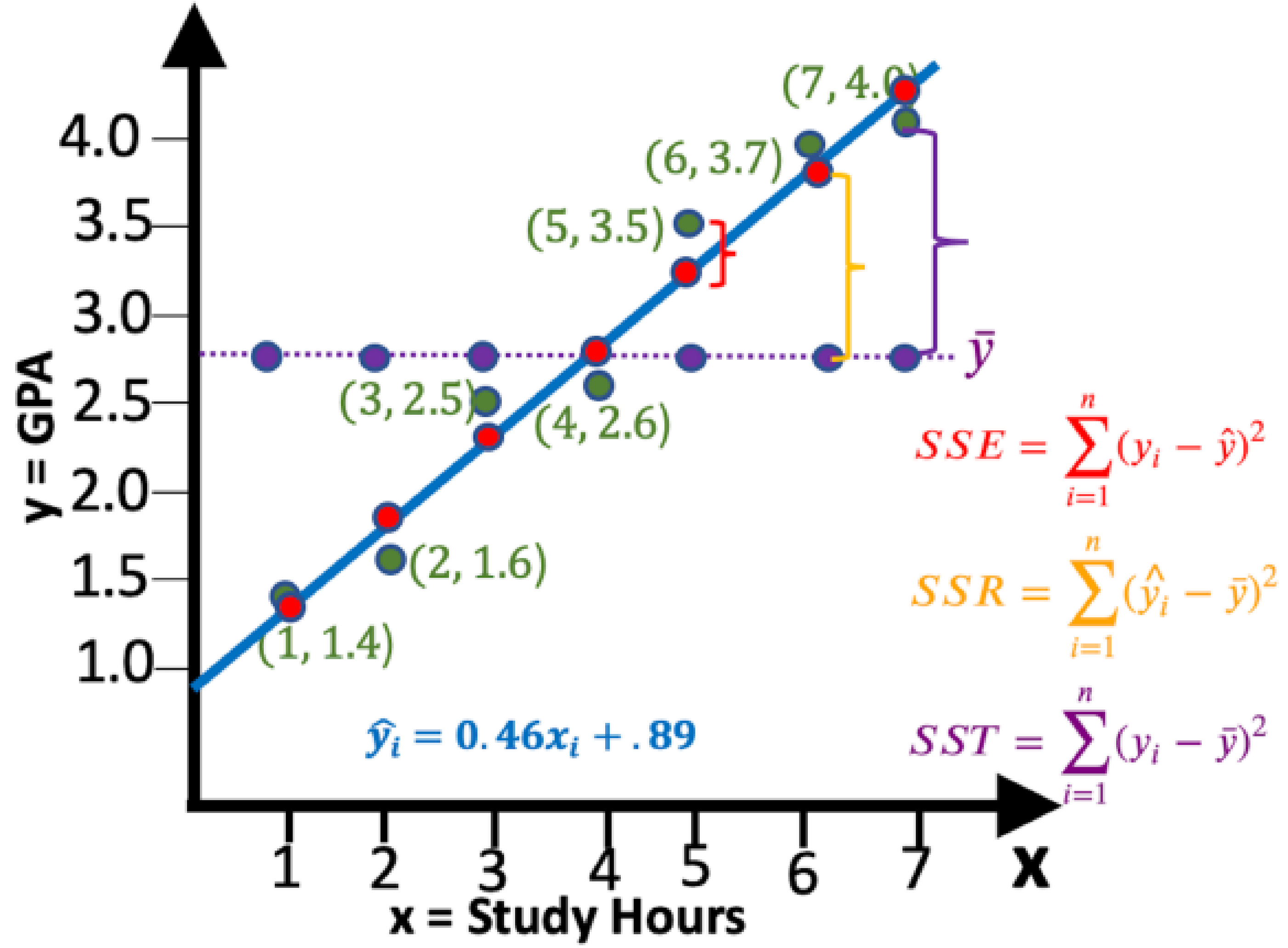
Derivation for MLR using OLS Method

- Taking the derivative of cost function and then solving it for zero to get the set of coefficients.
- After solving it you will get the following:

$$\beta = (X^T X)^{-1} X^T y$$

- Taking the derivative of cost function and then solving it for zero to get the set of coefficients becomes computationally expensive when the number of input features are hundreds or thousands.
- The Gauss Jordan Elimination technique of computing the inverse of a n x n matrix is $O(n^3)$. So, for 999 input features the size of X matrix will be , and the complexity of computing the inverse of this matrix will be 10^9
- Solution to this problem is Gradient Descent in which instead of using a closed form solution, we go slowly towards the answer using approximation technique.

Evaluate the Model



Thank You